TUTORIAL

Blocked alanine: optimization and a first scan

> Updated for EasyHybrid 3.2.0 (2026-10-26). The figures and numbers come from a run of this tutorial with the 3.2.0 development version.

The alanine dipeptide (blocked alanine, Ace-Ala-NMe, bALA) is the smallest molecule with the backbone of a protein: one ϕ/ψ pair and two peptide groups. In vacuum its lowest conformer, C7eq, is held by a hydrogen bond between the acetyl oxygen and the N-methyl amide hydrogen, closing a seven-membered ring.

In this tutorial you will:

1. create a system from a `.mol` file with the OPLS-AA force field;
2. optimize its geometry;
3. pick two atoms in the 3D view and scan their distance, breaking the hydrogen bond;
4. import the scan and read its energy profile in the PES Analysis window.

Everything runs in less than a minute. The structure is in the EasyHybrid folder: `examples/tutorials/01_bala/bALA_not_opt.mol`.

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1. Create the system

File → New (first button of the toolbar). Fill in the window as in Figure 1:

Field	Value
Name / Tag	`bALA` / `OPLS`
System Type	**OPLS**
Parameters	`protein` (pDynamo's OPLS-AA set for proteins, chosen by default)
Files	**Load Files →** `examples/tutorials/01_bala/bALA_not_opt.mol`
Create Folder	on: the working folder of the system, where the results are written

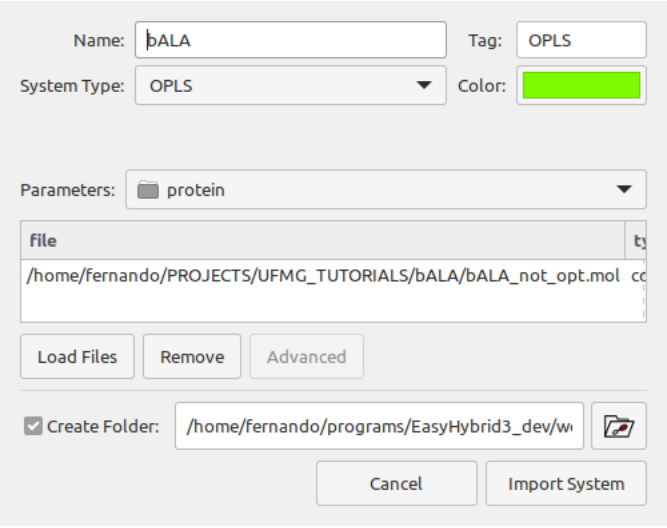


Figure 1. File → New with the OPLS force field.

Click **Import System**. The object list shows the system (`0 - bALA`) and its first object, `0_coords` , with one frame. The status bar at the bottom tells the force field (`force field: OPLS`) and the number of atoms (22).

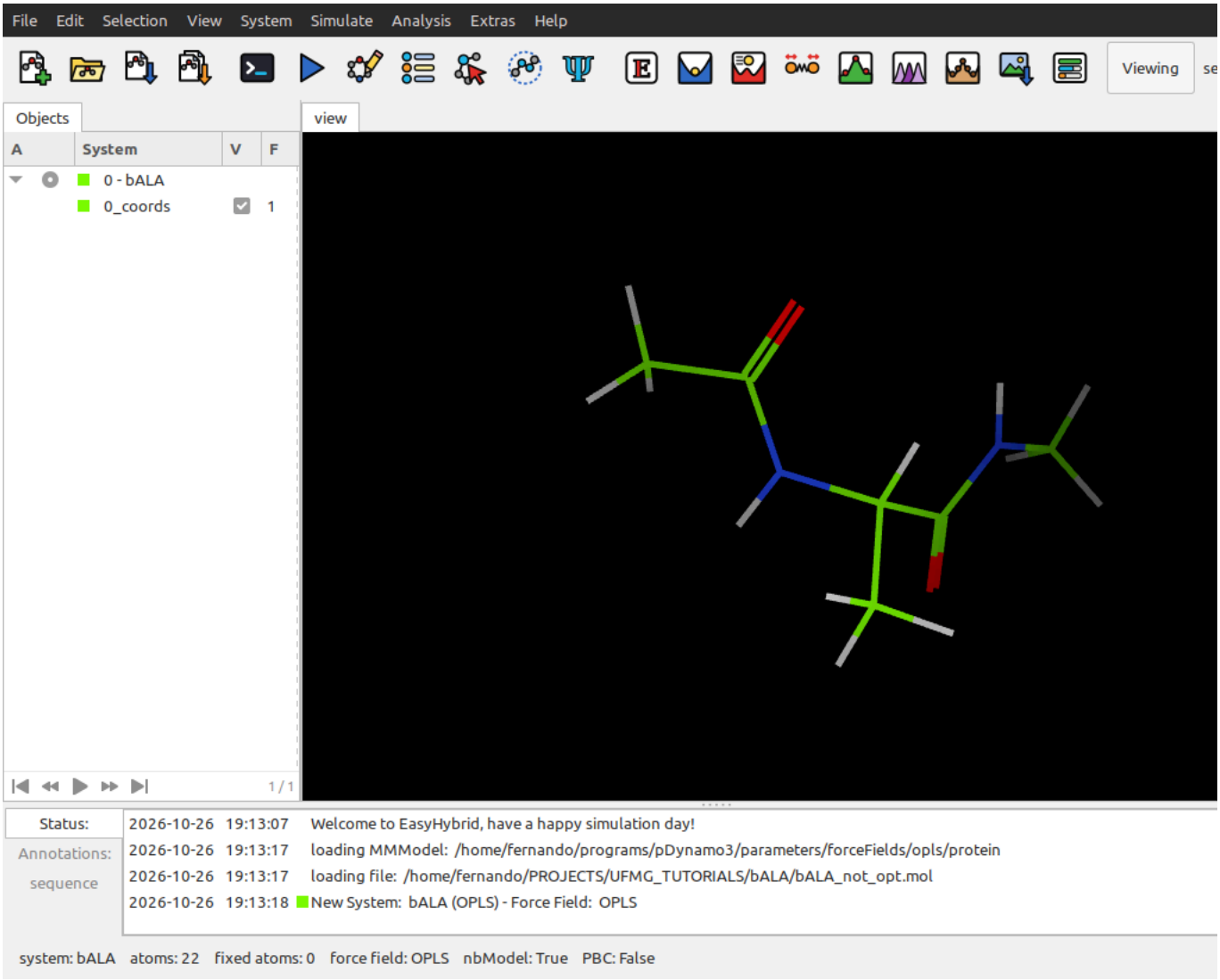


Figure 2. The new system in the main window.

The OPLS `protein` set recognizes the acetyl (ACE) and N-methyl (NME) caps by their atoms, so the `.mol` file needs no residue names.

2. Optimize the geometry

Simulate → Geometry Optimization (toolbar):

Field	Value
Coordinates	`0_coords`
Method	Conjugate Gradient
Maximum iterations	1000
RMS Grad. Tolerance	0.01 (kJ mol^{-1} Å^{-1})
Save Trajectory	off (only the final structure is kept)

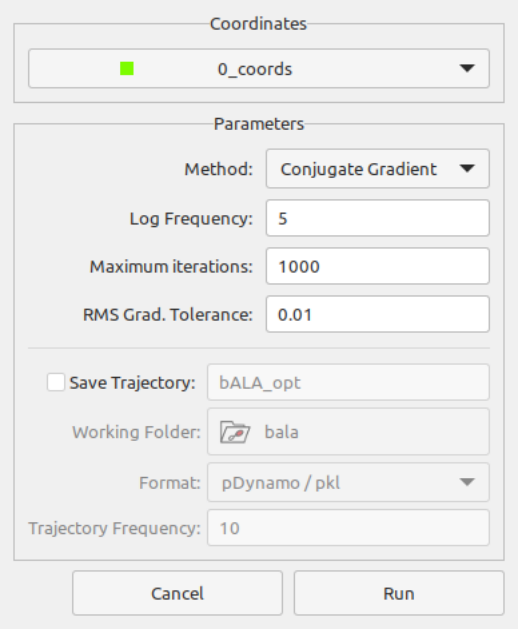


Figure 3. The Geometry Optimization window.

Click **Run**. The job takes about one second; when it ends, a new object with the optimized structure, `0_Geometry_Optimization_0`, appears in the object list. Its log is written in the working folder (double-click the job in the Process Manager to read it).

The optimized molecule is the C7eq conformer: $\phi = -80°$, $\psi = +68°$, and the O···H distance of the hydrogen bond is **2.06 Å**.

3. Pick the two atoms of the hydrogen bond

1. Hide `0_coords` (untick its **V** box) so that only the optimized structure is shown.
2. Switch the selection mode from **Viewing** to **Picking** (the button at the right of the toolbar).
3. Click the acetyl **oxygen** (O5) and then the amide **hydrogen** of the N-methyl group (H17). They are labeled `#1` and `#2`, and their distance is drawn between them (Figure 4).

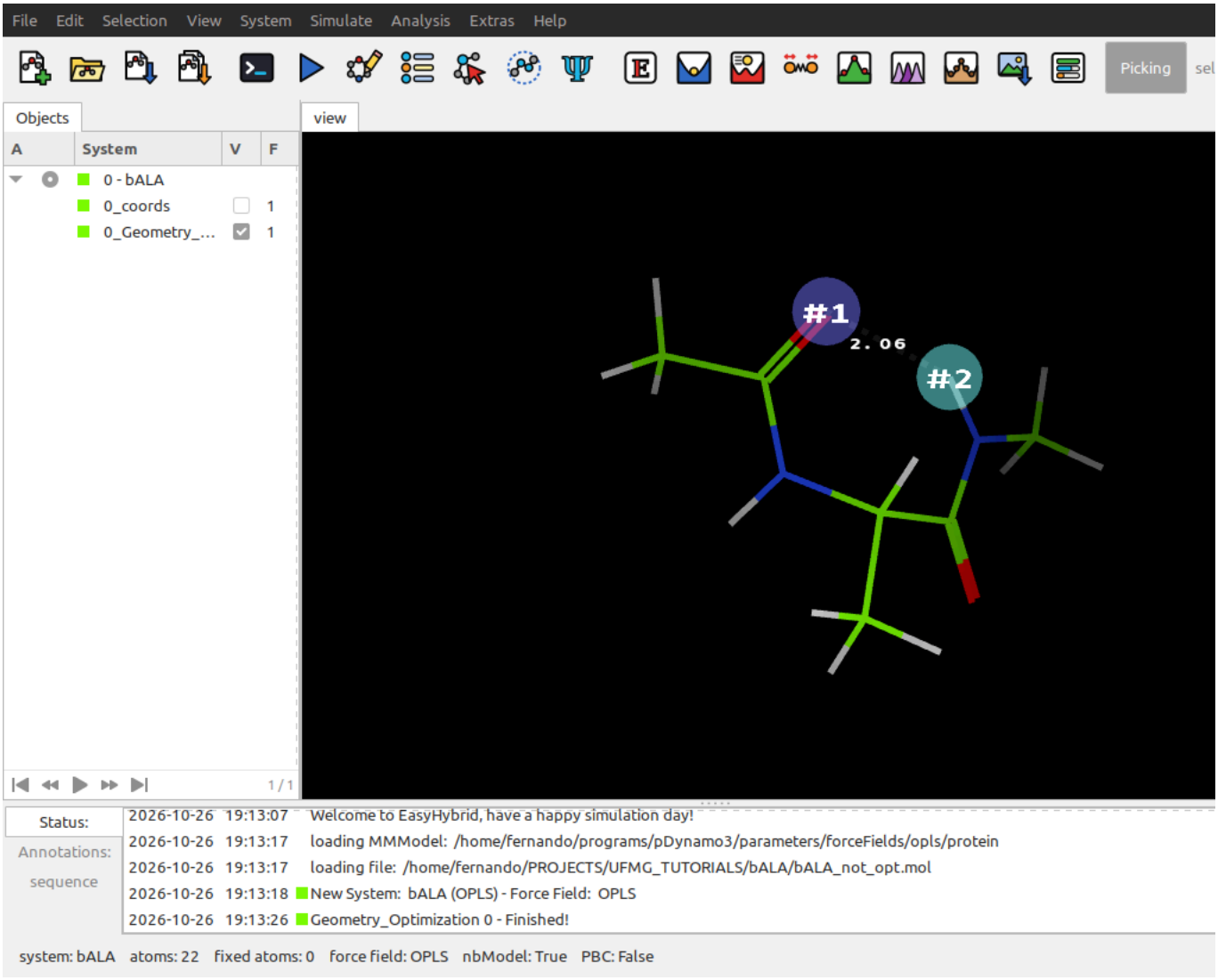


Figure 4. Atom #1 (O5) and atom #2 (H17) picked in the 3D view.

4. Scan the O···H distance

Simulate → Reaction Coordinate Scans (toolbar). In **Reaction Coordinate 1** keep the **Coordinate Type** *simple distance* and click **Import from Picking Selection**: the atom indices, names and the current distance (the **Initial Distance**) are filled in. Then:

Field	Value	Meaning
Step Size	0.2 Å	the distance grows 0.2 Å at each step
Number of Steps	20	20 points: from 2.06 to 5.85 Å
Force Constant	4000 kJ mol^{-1} Å^{-2}	holds the distance at each value
Coordinates	`0_Geometry_Optimization_0`	the scan starts from the optimized structure
Method	Conjugate Gradient	optimization at each point
Maximum iterations / RMS Grad. Tolerance	1000 / 0.1	
Trajectory Name	`bALA_scan_OH`	

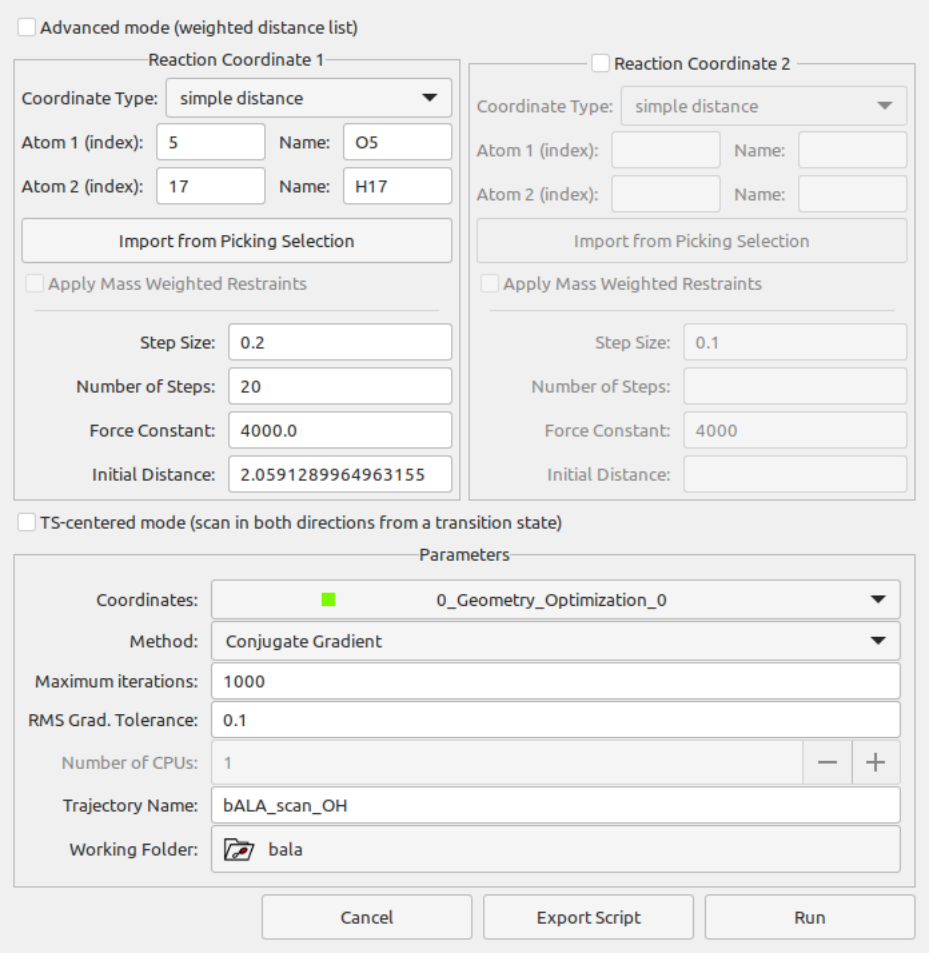


Figure 5. The scan of the O5···H17 distance.

Click **Run**. At each point a harmonic restraint holds the distance and the rest of the molecule is optimized. The run takes a few seconds and writes the folder `bALA_scan_OH.ptGeo` in the working folder: one structure per point (`frame0.pkl` … `frame19.pkl`) and `output.log` with the distances and energies.

5. Import the scan

When the job ends, the **Finished!** dialog appears. Click **Import...**: the **Import Data** window opens already filled in with the scan folder (format *pkl folder – pDynamo trajectory*) and its log (Figure 6). Click **Import**.

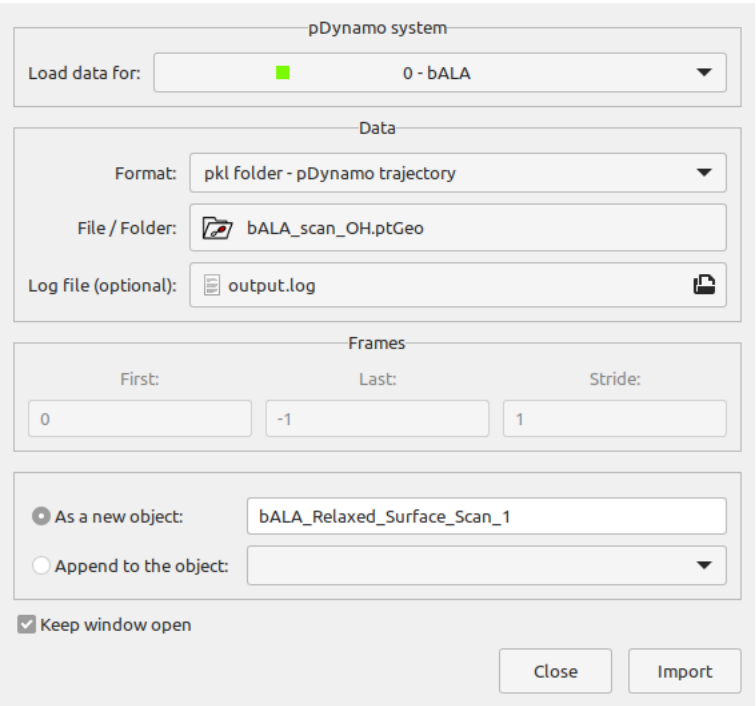


Figure 6. Import Data, pre-filled by the Finished! dialog.

A new object with 20 frames is added to the system. Play it with the trajectory buttons under the object list: the hydrogen bond opens and the backbone turns.

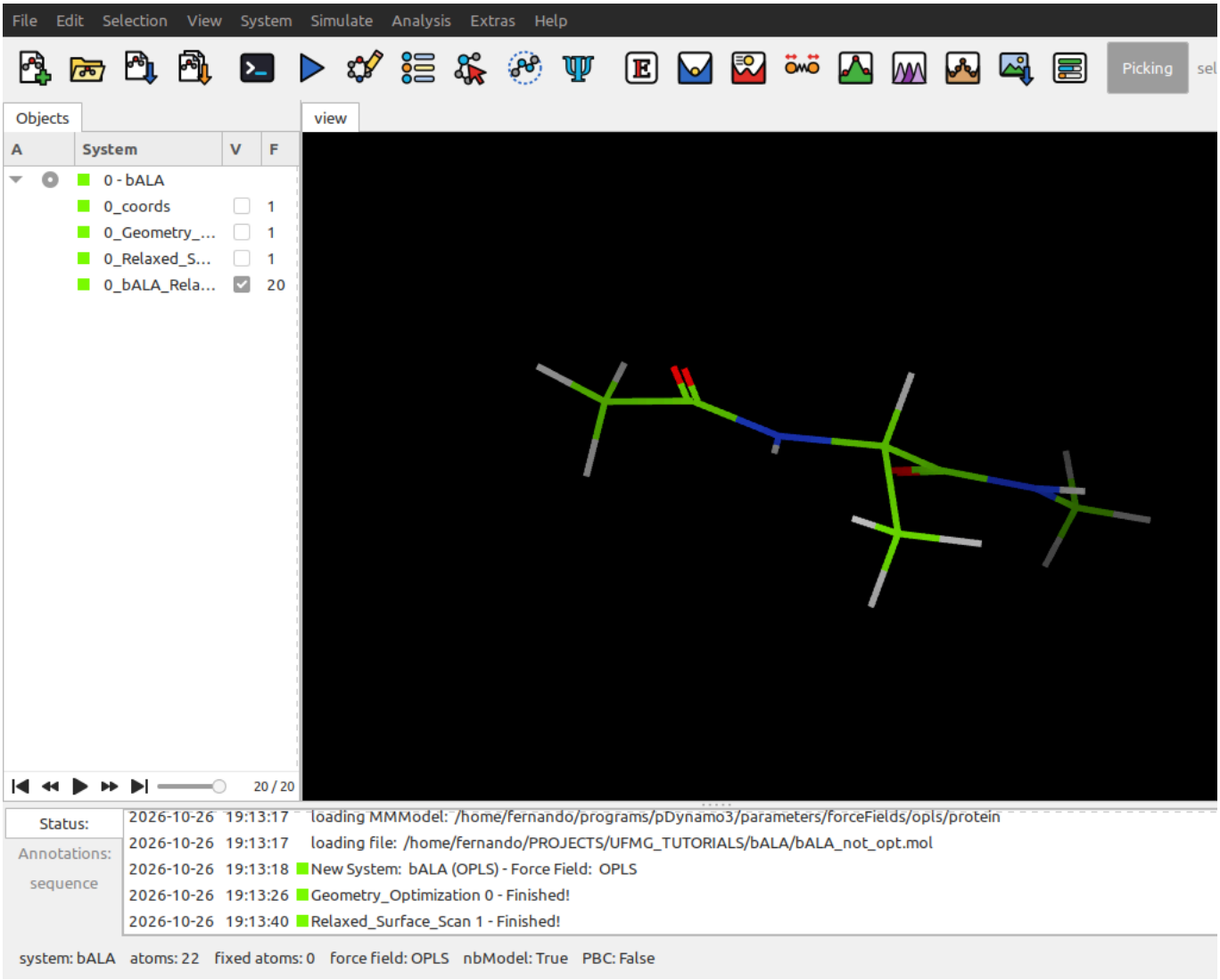


Figure 7. The last structure of the scan (O···H = 5.85 Å), an extended backbone.

6. The energy profile

Analysis → PES Analysis and choose the scan object in **Coordinates** (Figure 8). The energy is given relative to the first point.

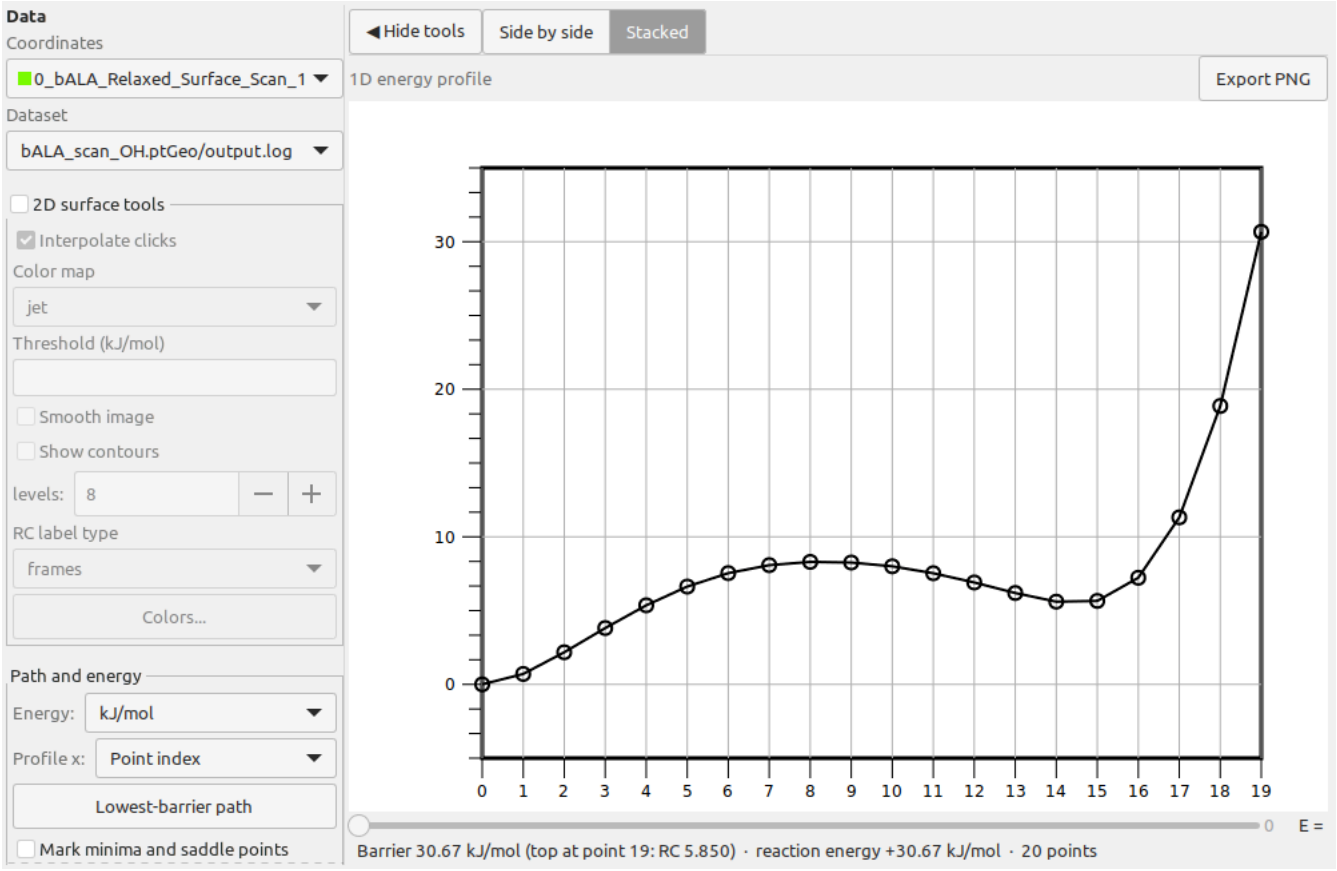


Figure 8. Energy along the O5···H17 distance.

Point	O···H (Å)	ϕ, ψ	E (kJ/mol)
0	2.06	−80°, +68° (C7eq)	0.0
8	3.66	−102°, +121°	8.3 (maximum)
14	4.86	−141°, +157° (C5 region)	5.6 (minimum)
19	5.85	−169°, −177°	30.7

Breaking the hydrogen bond costs about 8 kJ/mol; the molecule then falls into a second minimum near the C5 conformer (an extended backbone with a weaker N–H···O=C contact inside the same residue). Beyond 5 Å the distance can only grow by distorting the molecule, and the energy rises steeply: the end of the profile is the cost of stretching the restraint, not a chemical state.

Click a point of the profile, or move the slider under it, to show that structure in the 3D view.

Notes

- **The profile is not a free energy.** Each point is a minimum of the potential energy with the distance held; temperature and entropy are left out. The tutorial Umbrella sampling of blocked alanine computes the free energy along the same distance.
- **A relaxed scan follows one path.** If other coordinates jump between two points (here ϕ and ψ change smoothly), scan back from the end and compare.
- The barrier and the C5 minimum depend on the force field: OPLS-AA in vacuum. In water the C7eq hydrogen bond competes with water and the order of the conformers changes.

EasyHybrid 3.2.0 · built 2026-10-29 from `dev` @ `3539da4` · https://github.com/ferbachega/EasyHybrid3